

Mend-Amar Badral

 MendeBadra |  Mend-Amar Badral |  mendebadra.github.io
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Driven by a problem solving mindset and fueled by curiosity, I aspire to develop genuine solutions for the real-world, impactful problems.

SKILLS

Languages: English (C1 [IELTS 8.0](#) )
Programming: Python, Julia, C/C++, R, SQL
Tools: Linux, Git, Bash, PyTorch, Docker, Quarto, L^AT_EX

EDUCATION

Budapest University of Technology and Economics (BME) Sep, 2025 – June, 2027
M.Sc. in Computer Science Engineering
Budapest, Hungary
Supervisor: [Márton Vaitkus](#) 
Specialization: Data Science & Artificial Intelligence

National University of Mongolia (NUM) Sep, 2021 – June, 2025
B.Sc. in Applied Mathematics
Ulaanbaatar, Mongolia
GPA: 3.2/4.0
Supervisor: [Galtbayar Artbazar](#) 
Thesis: [Evaluating land degradation using image processing methods](#) 

EXPERIENCE

Center of Mathematics Application, NUM Oct, 2023 – June, 2025
Undergraduate Research Intern

- Developed [Python CLI scripts](#)  for multispectral drone image processing. Automated pipeline tasks and implemented multi-threading to reduce execution times by 80% for processing a total of 1800 multispectral drone imagery for 5 different sites.
- Configured and maintained secure remote access to lab servers via SSH and Tailscale VPN; managed access keys, automated system backups, and handled user environment provisioning.
- Architected a unified, documented data storage structure for research backups, directly improving data retrieval efficiency and establishing system-wide data management guidelines.
- Containerized data processing pipelines by deploying OpenDroneMap within Docker applications; configured environment dependencies and tuned parameters to ensure reproducible execution across CLI-driven Linux environments.
- Engaged in field survey data collection and multispectral measurements; gained first-hand insight into ecological data gathering to better align software pipelines with domain-specific research needs.

PROJECTS

[aiSim LiDAR point cloud sensor fusion for 3D Gaussian Splatting](#)

- Developed a Python scripts to colorize LiDAR point cloud measurements obtained from aiMotive's [aiSim](#)  vehicle simulation platform to train 3D Gaussian Splatting in [nerfstudio](#) .
- Engineered the transformation between 3D world coordinate frame to camera sensor coordinate frame, made the observations into a consistent coordinate convention by keeping track of each software stack's convention.

[Deep Learning Foundations and Concepts Book Examples Reproduction](#)

- Reproduced machine learning demonstrations from Bishop & Bishop's *Deep Learning Foundations and Concepts* using Julia notebooks and utilized innovative interactive sliders with reactivite programming environment using the [Pluto.jl](#)  package.

AWARDS

Stipendium Hungaricum Scholarship

Sep, 2025